

CSE 425: Internet of Things

7-Mark Exam Notes - Connecting Smart Objects

8 numbered points + each point 2–3 short lines + suitable for 7 marks.

1) 6LoWPAN — 8 Points HIGH PRIORITY

1. **6LoWPAN means Low-Power Wireless Personal Area Networks over IPv6.** It allows low-power wireless devices to communicate using Internet Protocol.
2. It is suitable for **small devices with limited processing capability.** Therefore, constrained IoT devices can connect to the Internet.
3. 6LoWPAN allows **IEEE 802.15.4 radios** to carry IPv6 communication. This makes it useful for low-power wireless IoT networks.
4. IPv6 uses a **128-bit address.** 6LoWPAN allows IEEE 802.15.4 devices to use these IPv6 addresses.
5. 6LoWPAN uses **header compression** techniques. Header compression reduces the size of the large IPv6 header.
6. IPv6 packets are **compressed and reformatted.** This allows them to fit into the smaller IEEE 802.15.4 packet format.
7. It provides Internet connectivity for low-power devices. Therefore, it is an important technology for IoT communication.
8. Main applications include **IoT, Smart Grid and M2M communication.** These applications normally use many low-power connected devices.

Memory: 128 → Compress → Fit → IoT

2) Addressing in 6LoWPAN — 8 Points

1. 6LoWPAN mainly supports **two types of addresses.** They are 64-bit Extended Address and 16-bit Short Address.
2. The **64-bit Extended Address** is globally unique. It can uniquely identify a device.
3. A 64-bit address is not limited to only one PAN. Therefore, it provides global identification.
4. The **16-bit Short Address** is smaller than the 64-bit address. It is used inside a particular PAN.
5. A 16-bit address is **PAN-specific.** It is assigned to a device by the PAN coordinator.
6. IEEE 802.15.4 does not directly support **IPv6 multicast.** Therefore, multicast communication needs another method.
7. IPv6 packets can be carried as **link-layer broadcast frames.** This allows IPv6 traffic to operate over IEEE 802.15.4.

8. The easiest difference is that **64-bit is global and 16-bit is PAN-specific**. This is the main idea of 6LoWPAN addressing.

Memory: 64 = Global, 16 = PAN

3) 6LoWPAN Header Types — 8 Points

1. 6LoWPAN uses several header types for communication. The important ones are Dispatch, Mesh Addressing and Fragmentation headers.
2. The **Dispatch Header** is used at the beginning of communication. It helps identify the type of the next header.
3. The Dispatch field is **6 bits long**. It identifies which header or protocol comes next.
4. A **Type Specific Header** may follow the Dispatch Header. Its format depends on the value of the Dispatch Header.
5. The **Mesh Addressing Header** is used for mesh forwarding. It includes Originator Address, Final Address and Hops Left.
6. The **Hops Left** value decreases at every forwarding node. This shows how many more hops the packet can travel.
7. The **Fragmentation Header** is used when a large packet must be divided. The first fragment contains Datagram Size and Datagram Tag.
8. Later fragments also contain a **Datagram Offset**. This helps the receiver put the fragments back in the correct order.

Memory: DMF = Dispatch → Mesh → Fragmentation

4) LOADng Routing — 8 Points HIGH PRIORITY

1. **LOADng is derived from AODV** routing protocol. It was extended for use in IoT networks.
2. LOADng discovers a route when the route is required. Therefore, it works as an on-demand routing approach.
3. The originator generates a **Route Request (RREQ)**. The RREQ is used to discover a route to the destination.
4. The RREQ is forwarded through different LOADng routers. It continues until it reaches the destination.
5. The destination generates a **Route Reply (RREP)**. The RREP is sent hop-by-hop back toward the originator.
6. If a route becomes broken, a **Route Error (RERR)** is generated. It informs the originator about the route failure.

7. LOADng supports **optimized flooding**. This reduces the overhead caused by RREQ generation and flooding.
8. Only the **destination can reply to an RREQ**. Intermediate LOADng routers are not allowed to reply.

Memory: RREQ → RREP → RERR

5) RPL Routing — 8 Points HIGH PRIORITY

1. **RPL is a Distance Vector IPv6 routing protocol**. It is designed for Low-Power and Lossy Networks or LLNs.
2. RPL maintains routing topology using **low-rate beaconing**. Under normal conditions, beacon messages are sent at a low rate.
3. The beaconing rate increases when an inconsistency is detected. For example, this can happen when a node or link becomes unavailable.
4. Routing information can be included in the **datagram itself**. Therefore, routing information travels with the packet.
5. **Proactive operation** is used to maintain routing topology. It keeps the routing structure ready before a problem occurs.
6. **Reactive operation** resolves routing inconsistencies. It reacts when a problem appears in the network.
7. RPL supports **confidentiality, integrity, loop detection and data-path validation**. These features improve routing security and correctness.
8. RPL tries to minimize **energy and latency** and satisfy network constraints. Constraints may include node power and bandwidth.

Memory: P = Preserve, R = Repair, ELC = Energy-Latency-Constraints

6) LOADng vs RPL — 8 Points

1. LOADng is derived from **AODV**, while RPL is a **Distance Vector IPv6 protocol**. Both are related to routing in IoT networks.
2. LOADng discovers routes mainly **when they are needed**. RPL mainly maintains a routing topology continuously.
3. LOADng uses **RREQ** for route discovery. The source sends an RREQ when it needs a path.
4. The LOADng destination sends an **RREP** back to the source. RPL instead uses beaconing to maintain topology.
5. LOADng uses **RERR** when a route becomes broken. RPL increases beaconing when it detects an inconsistency.

6. Only the destination can answer an RREQ in LOADng. Intermediate routers are not permitted to send the reply.
7. RPL has both **proactive and reactive** behavior. Proactive maintains topology and reactive resolves inconsistencies.
8. RPL is specially designed for **Low-Power and Lossy Networks**. LOADng mainly provides lightweight on-demand route discovery.

Memory: LOADng = Find route when needed RPL = Maintain route

7) RFID — 8 Points HIGH PRIORITY

1. **RFID stands for Radio-Frequency Identification.** It uses radio-frequency technology to identify objects.
2. Information is digitally stored inside an **RFID tag**. An RFID reader can read this stored information.
3. RFID is somewhat similar to barcode technology. However, RFID uses radio communication.
4. Information read from the tag can be stored in a **database**. This makes automatic tracking and identification possible.
5. RFID does **not require direct line-of-sight**. This is an important advantage over traditional barcodes and QR codes.
6. An RFID tag mainly contains an **integrated circuit and an antenna**. Protective material can also cover and protect the tag.
7. RFID tags may be **Passive or Active**. Passive RFID tags are the most widely used.
8. Passive tags receive power from the reader, while Active tags have their own power supply. This is the main difference between the two tag types.

Memory: RFID = Radio + Tag + Reader

8) Passive vs Active RFID — 8 Points

1. RFID tags can mainly be divided into **Passive and Active tags**. Their main difference is the source of power.
2. A **Passive tag has no internal power supply**. It cannot operate independently without a reader.
3. A Passive tag receives power **inductively from the reader**. The reader activates the tag before it sends information.
4. Passive RFID tags are the **most widely used** type. They are common in many RFID applications.
5. An **Active RFID tag has its own power supply**. Therefore, it does not fully depend on reader power.

6. Both Passive and Active tags store digitally encoded information. An RFID reader can read this information.
7. RFID tags contain an **integrated circuit and an antenna**. These are the main physical parts of a tag.
8. The simple difference is: **Passive = reader-powered, Active = self-powered**. This is the easiest way to remember the comparison.

Memory: Passive = No own power | Active = Own power

9) RFID Working Principle — 8 Points

1. RFID is related to **AIDC — Automatic Identification and Data Capture**. AIDC automatically identifies objects and collects their data.
2. AIDC performs **object identification**. It recognizes the object without much human involvement.
3. It also performs **data collection and mapping to computer systems**. The process requires little or no human intervention.
4. RFID uses **radio waves** to perform AIDC functions. Radio communication is used between the tag and reader.
5. The main RFID system components are **Tag, Reader and Antenna**. Together they form the basic RFID system.
6. The RFID tag is attached to the object to be identified. It stores digitally encoded information.
7. The RFID reader receives information from the tag. The information can then be sent to a computer or database.
8. The simple working flow is: **Tag → Radio Waves → Reader → Computer/Database**. This allows automatic identification and tracking.

Memory: Tag → Reader → Database

10) RFID Applications — 8 Points

1. **Inventory Management:** RFID can automatically identify and manage products in a store or warehouse.
2. **Asset Tracking:** It can track valuable equipment and other organizational assets.
3. **Personnel Tracking:** RFID can be used to identify or track employees and authorized persons.
4. **Access Control:** RFID cards can control entry into restricted areas.
5. **ID Badging:** RFID can be used in employee or identification badges.
6. **Supply Chain Management:** Goods can be tracked from production to distribution.

7. **Counterfeit Prevention:** RFID can help detect fake products, especially in the pharmaceutical industry.
 8. Overall, RFID is useful for **automatic identification, security and tracking**. Therefore, it has many industrial and commercial applications.
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11) ZigBee / IEEE 802.15.4 — 8 Points HIGH PRIORITY

1. ZigBee uses **IEEE 802.15.4 at the PHY and MAC layers**. Higher ZigBee layers provide network, security and application functions.
2. The ZigBee network layer uses **AODV routing**. AODV means Ad Hoc On-Demand Distance Vector.
3. An important PHY frequency is **2.4 GHz**. It provides 16 channels and a data rate of 250 kbps.
4. Other options are **915 MHz at 40 kbps and 868 MHz at 20 kbps**. Different frequencies provide different data rates.
5. The MAC layer performs **network beaconing and PAN association/disassociation**. Beacons can help new devices join the network.
6. The MAC layer also provides **device security and reliable link communication**. These functions support reliable communication between devices.
7. Four MAC frame types are **Data, Beacon, Acknowledgement and MAC Command**. They handle data transfer, beaconing, confirmation and control.
8. IEEE 802.15.4 supports different topologies and uses **AES encryption** for security. AES provides secure communication and data integrity.

Memory: DBAM = Data, Beacon, ACK, MAC

12) IEEE 802.11ah — 8 Points HIGH PRIORITY

1. IEEE 802.11ah is a **sub-GHz Wi-Fi technology** for IoT applications. It was developed to overcome some limitations of normal Wi-Fi.
2. Normal Wi-Fi has problems with low-power operation and large device populations. IEEE 802.11ah was designed to improve these areas.
3. IEEE 802.11ah operates in **unlicensed sub-GHz bands**. The slide also identifies it as Wi-Fi HaLow.
4. It can support up to **8192 devices per access point**. Therefore, it is suitable for large IoT deployments.
5. Its **MAC header is shortened** for efficient communication. This helps reduce communication overhead.

6. **RAW — Restricted Access Window** reduces simultaneous transmissions. It reduces collisions and can save battery power.
7. **TWT — Target Wake Time** controls when a device wakes and accesses the network. Devices can remain in a low-power state until their scheduled time.
8. IEEE 802.11ah mainly uses **star topology**, with relay support for longer range. Client devices can help relay data.

Memory: RAW = Reduce Collision | TWT = Time To Wake

13) IEEE 1901.2a / NB-PLC — 8 Points

1. IEEE 1901.2a is a **wired communication technology**. This makes it different from many wireless IoT technologies.
2. It is a standard for **Narrowband Power Line Communication (NB-PLC)**. It uses a narrowband spectrum for communication.
3. NB-PLC uses the same wires that carry **electrical power**. Therefore, power lines also act as the communication medium.
4. It supports **low-power, long-range and interference-resistant communication**. These characteristics are useful in utility networks.
5. One important application is **Smart Metering**. It can automate electric, gas and water meter readings.
6. It is also used for **Distribution Automation and Public Lighting**. Power-grid devices can be monitored and controlled.
7. Other applications include **EV charging stations and microgrids**. It is useful in modern energy infrastructure.
8. It can also be used in **renewable energy systems**. Examples include solar, wind, hydroelectric and geothermal applications.

Memory: Power Line = Communication Line

14) IEEE 802.15.4e — 8 Points

1. IEEE 802.15.4e is an **amendment to IEEE 802.15.4-2011**. It expands the features of the MAC layer.
2. Its main purpose is to improve limitations of the original standard. The improvements mainly focus on MAC-layer performance.
3. It improves **MAC reliability**. This helps make wireless communication more dependable.
4. It addresses the problem of **unbounded latency**. This is useful for applications that need better timing behavior.

5. It also improves operation under **multipath fading**. This helps wireless networks work better in difficult environments.
6. IEEE 802.15.4e provides general **MAC-layer enhancements**. These enhancements extend the capabilities of IEEE 802.15.4.
7. It is useful in **factory and process automation**. These applications need reliable industrial communication.
8. It is also useful in **Smart Grid applications**. Therefore, IEEE 802.15.4e is important for industrial and utility IoT.

Memory: e = Enhanced MAC

Final exam priority

Must study first: 6LoWPAN → LOADng → RPL → RFID → Zig-Bee/802.15.4 → IEEE 802.11ah.